

Single neurons detect spatiotemporal activity transitions through STP and EI imbalance

Supplementary Methods: Model Parameters

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1 Chemical model for STP synapses

1.1 GABA Boutons

Table 1.1A: Reactions for GABA boutons

Reaction	K_f	K_b
$2 \text{ Ca} \rightleftharpoons 2 \text{ Ca}_{\text{ext}}$	$21.238 \mu\text{M}^{-1} \cdot \text{s}^{-1}$	$6.9463 \mu\text{M}^{-1} \cdot \text{s}^{-1}$
$\text{GABA} \rightleftharpoons 2 \text{ Ca} + \text{vesicle_pool}$	38544 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$\text{vesicle_pool} \rightleftharpoons \text{RR_pool}$	2.6639 s^{-1}	2.6639 s^{-1}
$2 \text{ Ca} + \text{Docked} \rightleftharpoons \text{GABA}$	$0.52216 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	0 s^{-1}
$2 \text{ Ca} + \text{RR_pool} \rightleftharpoons \text{Ca}_{\text{RR}}$	$16.017 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	3.349 s^{-1}
$\text{Ca}_{\text{RR}} \rightleftharpoons 2 \text{ Ca} + \text{Docked}$	341.59 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$\text{Docked} \rightleftharpoons \text{RR_pool}$	335.88 s^{-1}	0 s^{-1}
$4 \text{ Ca} + \text{Buffer} \rightleftharpoons \text{Ca}_4.\text{Buffer}$	$335.88 \mu\text{M}^{-4} \cdot \text{s}^{-1}$	1000 s^{-1}

Table 1.1B: Pools for GABA boutons

Name	Initial Concentration	Buffered
Ca	$0.08 \mu\text{M}$	0
Ca_{ext}	$0.08 \mu\text{M}$	1
RR_pool	$3 \mu\text{M}$	0
vesicle_pool	$3.1699 \mu\text{M}$	1
Docked	$0 \mu\text{M}$	0
Ca_{RR}	$0 \mu\text{M}$	0
GABA	$0 \mu\text{M}$	0
Buffer	$3.4727 \mu\text{M}$	0
$\text{Ca}_4.\text{Buffer}$	$0 \mu\text{M}$	0

1.2 Glutamate Boutons

Table 1.2A: Reactions for Glu boutons

Reaction	K_f	K_b
$2 \text{ Ca} \rightleftharpoons 2 \text{ Ca}_{\text{ext}}$	$47.118 \mu\text{M}^{-1} \cdot \text{s}^{-1}$	$15.889 \mu\text{M}^{-1} \cdot \text{s}^{-1}$
$\text{glu} \rightleftharpoons 2 \text{ Ca} + \text{vesicle_pool}$	74957 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$\text{vesicle_pool} \rightleftharpoons \text{RR_pool}$	2.6147 s^{-1}	2.6147 s^{-1}
$2 \text{ Ca} + \text{Docked} \rightleftharpoons \text{glu}$	$0.62741 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	0 s^{-1}
$2 \text{ Ca} + \text{RR_pool} \rightleftharpoons \text{Ca}_{\text{RR}}$	$0.99392 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	0.17877 s^{-1}
$\text{Ca}_{\text{RR}} \rightleftharpoons 2 \text{ Ca} + \text{Docked}$	1986.1 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$\text{Docked} \rightleftharpoons \text{RR_pool}$	127.73 s^{-1}	0 s^{-1}
$4 \text{ Ca} + \text{Buffer} \rightleftharpoons \text{Ca}_4.\text{Buffer}$	$127.73 \mu\text{M}^{-4} \cdot \text{s}^{-1}$	1000 s^{-1}

Table 1.2B: Pools for Glutamate boutons

Name	Initial Concentration	Buffered
Ca	$0.08 \mu\text{M}$	0
Ca_{ext}	$0.08 \mu\text{M}$	1
RR_pool	$0.3 \mu\text{M}$	0
vesicle_pool	$0.872 \mu\text{M}$	1
glu	$0 \mu\text{M}$	0
Docked	$0 \mu\text{M}$	0
Ca_{RR}	$0 \mu\text{M}$	0
Buffer	$4.0953 \mu\text{M}$	0
$\text{Ca}_4.\text{Buffer}$	$0 \mu\text{M}$	0

2 Chemical model for non-STP synapses

2.1 GABA boutons

Table 2.1A: Reactions for GABA boutons

Reaction	K_f	K_b
$2 \text{ Ca} \rightleftharpoons 2 \text{ Ca}_{\text{ext}}$	$21.238 \mu\text{M}^{-1} \cdot \text{s}^{-1}$	$6.9463 \mu\text{M}^{-1} \cdot \text{s}^{-1}$
$\text{GABA} \rightleftharpoons 2 \text{ Ca} + \text{vesicle_pool}$	38544 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$2 \text{ Ca} + \text{Docked} \rightleftharpoons \text{GABA}$	$0.52216 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	0 s^{-1}

Table 2.1B: Pools for GABA boutons

Name	Initial Concentration	Buffered
Ca	$0.08 \mu\text{M}$	0
Ca_{ext}	$0.08 \mu\text{M}$	1
vesicle_pool	$3.1699 \mu\text{M}$	1
Docked	$0.5 \mu\text{M}$	1
GABA	$0 \mu\text{M}$	0

2.2 Glutamate boutons

Table 2.2A: Reactions for glu boutons

Reaction	K_f	K_b
$2 \text{ Ca} \rightleftharpoons 2 \text{ Ca}_{\text{ext}}$	$47.118 \mu\text{M}^{-1} \cdot \text{s}^{-1}$	$15.889 \mu\text{M}^{-1} \cdot \text{s}^{-1}$
$\text{glu} \rightleftharpoons 2 \text{ Ca} + \text{vesicle_pool}$	74957 s^{-1}	$0 \mu\text{M}^{-2} \cdot \text{s}^{-1}$
$2 \text{ Ca} + \text{Docked} \rightleftharpoons \text{glu}$	$0.62741 \mu\text{M}^{-2} \cdot \text{s}^{-1}$	0 s^{-1}

Table 2.2B: Pools for glu boutons

Name	Initial Concentration	Buffered
Ca	$0.08 \mu\text{M}$	0
Ca_{ext}	$0.08 \mu\text{M}$	1
vesicle_pool	$0.872 \mu\text{M}$	1
glu	$0 \mu\text{M}$	0
Docked	$0.225 \mu\text{M}$	1

3 Electrical Model

The voltage V is in mV and referenced to the resting potential. Time is in milliseconds (ms).

3.1 Ion Channel Definitions

These definitions are mostly from Traub et al, 1991.

Ion	Conductance	Reversal	Gate	α	β
Na	$g_{\text{Na}} = g_{\text{max_Na}} \cdot m^2 h$	$E_{\text{Na}} = 115 \text{ mV}$	m-gate	$\frac{0.32(13.1 - V)}{\exp\left(\frac{13.1 - V}{4}\right) - 1}$	$\frac{0.28(V - 40.1)}{\exp\left(\frac{V - 40.1}{5}\right) - 1}$
			h-gate	$0.128 \exp\left(\frac{17 - V}{18}\right)$	$\frac{4}{1 + \exp\left(\frac{40 - V}{5}\right)}$
K	$g_{\text{KDR}} = g_{\text{max_KDR}} \cdot n$	$E_{\text{KDR}} = -15 \text{ mV}$	n-gate	$\frac{0.016(35.1 - V)}{\exp\left(\frac{35.1 - V}{5}\right) - 1}$	$0.25 \exp\left(\frac{20 - V}{40}\right)$

3.2 Receptor-gated Ion Channel Conductances

Glutamate Receptor: AMPA

$$g_{\text{GluR}} = \frac{A \cdot g_{\text{maxGluR}}}{\tau_1 - \tau_2} \left(\exp\left(\frac{-t}{\tau_1}\right) - \exp\left(\frac{-t}{\tau_2}\right) \right)$$

where: A = normalization constant such that $g_{\text{GluR}} = g_{\text{maxGluR}}$ at peak. $\tau_1 = 2, \tau_2 = 9$

GABA Receptor

$$g_{\text{GABAR}} = \frac{A \cdot g_{\text{maxGABAR}}}{\tau_1 - \tau_2} \left(\exp\left(\frac{-t}{\tau_1}\right) - \exp\left(\frac{-t}{\tau_2}\right) \right)$$

where: A = normalization constant such that $g_{\text{GABAR}} = g_{\text{maxGABAR}}$ at peak. $\tau_1 = 4, \tau_2 = 9$

Glutamate Receptor: NMDA

$$g_{\text{NMDAR}} = \frac{g_{\text{maxNMDAR}}}{\tau} \exp\left(\frac{-t}{\tau}\right) \frac{K_{\text{Mg}}}{K_{\text{Mg}} + [\text{Mg}]}$$

where: $K_{\text{Mg}} = \frac{\gamma \cdot \exp(V - E_{\text{rest}})}{\eta}$, $\tau = 20$, $\gamma = 0.28$, $\eta = 62$

$$I_{\text{NMDARCa}} = g_{\text{NMDAR}} \cdot \text{Ca}_{\text{frac}} \cdot \ln\left(\frac{[\text{Ca}_{\text{out}}]}{[\text{Ca}_{\text{in}}]}\right) \cdot V \cdot \frac{[\text{Ca}_{\text{in}}] - \varphi[\text{Ca}_{\text{out}}]}{(1 - \varphi)([\text{Ca}_{\text{in}}] - [\text{Ca}_{\text{out}}])}$$

Where: $F = 96485 \text{ sA/mol}$, $z = 2$, $R = 8.314 \text{ J/(K} \cdot \text{mol)}$, $T = 300 \text{ K}$, $\varphi = \exp(-VFz/RT)$, Ca_{frac} = fraction of current carried at 0 mV by Ca = 0.02, $[\text{Ca}_{\text{out}}] = 1.5 \text{ mM}$, $[\text{Ca}_{\text{in}}] = 0.08 \text{ } \mu\text{M}$

3.3 Calcium Pools

$$\frac{d[\text{Ca}]}{dt} = \phi(I_{\text{Ca}} + \text{Ca}_{\text{NMDA}}) - \frac{[\text{Ca}]}{13.33}$$

where: $\phi =$

3.4 Passive Properties

$$R_M = 0.2 \, \Omega \cdot \text{m}^2, \quad R_A = 1.0 \, \Omega \cdot \text{m}, \quad C_M = 0.01 \, \text{F}/\text{m}^2, \quad E_{\text{rest}} = -70 \, \text{mV}$$

3.5 Channel Distributions

Channel	Zone	Distribution (Gmax, S/m ²)
Na	Soma	600
K_DR	Soma	360
GABAR	Dendrite	50
GluR	Spine Heads	200
NMDAR	Spine Heads	80

References:

Traub RD, Wong RK, Miles R, Michelson H. A model of a CA3 hippocampal pyramidal neuron incorporating voltage-clamp data on intrinsic conductances. J Neurophysiol. 1991 Aug;66(2):635-50. doi: 10.1152/jn.1991.66.2.635. PMID: 1663538.